

Program: Diploma in Communication and ComputerNetworking	
Course Code: 3323	Course Title: Object Oriented Programming using JAVA
Semester: 3	Credits: 3
Course Category: Programme core course	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- Understand basic object oriented concepts of JAVA, Java run time environment, development platform etc.
- Identify the data types and variables in Java, the operators and different Control Structures in Java
- Understand the concept of classes, objects and methods, Input / Output operations in Java.
- Understand the concept of Inheritance, Packages and Interfaces, Exception Handling.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	CognitiveLevel
CO1	Have an overview of Java	12	Understanding
CO2	To understand the concepts of data type, variable and control structures in Java	11	Applying
CO3	To understand the concept of classes, objects and methods	10	Applying
CO4	To understand the Input / Output operations in Java	10	Applying
	SeriesTest	2	

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	1	2		2		
CO3	3	2	1		2		
CO4	3	2	1				

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Have an overview of Java		
M1.01	To understand basic object oriented concept	4	Understanding
M1.02	To understand Java run time environment, development platform	4	Understanding
M1.03	To Understand Java Virtual Machine, Java Compiler, Byte Code	4	Understanding
Contents: Overview of Java: An overview of Java, Features of Java, Basic Object Oriented concepts, Java programming Environment and Runtime Environment, Development Platforms -Standard, Enterprise. Java Virtual Machine (JVM), Java compiler, Byte code.			
CO2	To understand the concepts of data type, variable and control structures in Java		
M2.01	To Identify the data types and variables in Java	4	Understanding
M2.02	To describe the operators and different Control Structures in Java	6	Applying
	Series Test -I	1	

Contents: Core Java Fundamentals: Core Java Fundamentals: Data types, variables, Type Conversion and Casting, Arrays, Operators - Arithmetic Operators, Bitwise Operators, Relational Operators, Boolean Logical Operators, Assignment Operator, Conditional (Ternary) Operator, and Operator Precedence. Control Statements - Selection Statements, Iteration Statements and Jump Statements.			
CO3	To understand the concept of classes, objects and methods		
M3.01	To understand Creation and implementation of classes	6	Understanding
M3.02	To understand the Input / Output operations	4	Understanding
Contents: Understand the concept of classes, objects and methods -Creation and implementation of classes, objects in Java,Constructors, Method Overloading, Static Members, Final Variables- understand the Input / Output operations in Java-Reading data from console, Writing data to console			
CO4	Inheritance, Packages and Interfaces, Exception Handling		
M4.01	To demonstrate Inheritance, Packages and Interfaces.	7	Understanding
M4.02	To understand Exception Handling in JAVA	6	Applying
	SeriesTest–II	1	
Contents: Inheritance, Packages and Interfaces -Base class and derived class,Method Overriding,Abstract Classes and Methods,Use of final keyword,Concept of interfaces,Concept of packages ,Exceptions ,Exception handling keyword			

Text/Reference

T/R	Book Title / Author
T1	Herbert Schildt and Dale Skrien, "Java Fundamentals – A comprehensive Introduction", McGraw Hill, 1st Edition, 2013.
T2	Herbert Schildt, "Java the complete reference", McGraw Hill, Osborne, 7th Edition, 2011
T3	T.Budd, "Understanding Object- Oriented Programming with Java", Pearson Education, Updated Edition (New Java 2 Coverage), 1999
R1	P.J.Dietel and H.M.Dietel , "Java How to program", Prentice Hall, 6th Edition, 2005

R2	P.Radha Krishna , “Object Oriented programming through Java”, CRC Press, 1 st Edition, 2007
R3	S.Malhotra and S. Choudhary, “Programming in Java”, Oxford University Press, 2nd Edition, 2014

OnlineResources

Sl No	WebsiteLink
1	http://java.sun.com 2. http://www.oracle.com/technetwork/java/index.html)
2	http://java.sun.com/javase
3	http://www.oracle.com/technetwork/java/javase/overview/index.html 5.
4	http://download.oracle.com/javase/7/docs/api/index.html
5	http://docs.oracle.com/javase/tutorial/
6	iiti.ac.in/people/~tanimad/JavaTheCompleteReference.pdf
7	https://www.codejava.net/books/4-best-free-java-e-books-for-beginners